

FedRansom: Federated Learning Based Privacy-Preserving Ransomware Detection Using Heterogeneous Multi-Modal Architecture

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Abstract—Ransomware attacks continue to inflict devastating damage on organizations worldwide, while data privacy regulations prevent collaborative defense through centralized machine learning. This paper presents FedRansom, a heterogeneous federated learning framework for privacy-preserving ransomware detection. The system integrates two complementary modalities: a MobileNetV2 Convolutional Neural Network (CNN) trained on Malimg and MaleBin malware visualization datasets for 60-class family classification, and a Multi-Layer Perceptron (MLP) trained on EMBER 2018 and EMBER 2024 static Portable Executable (PE) file features for binary malware detection. Manual Federated Averaging (FedAvg) aggregates client models across 10 communication rounds, with Federated Model Distillation (FedMD) enabling cross-architecture knowledge transfer between heterogeneous CNN and MLP global models. Evaluation across 1.3 million samples demonstrates that the federated image model achieves 92.32% accuracy, surpassing the centralized baseline by 3.04%, while the EMBER 2024 federated model achieves 94.53%, exceeding centralized performance. Under non-IID data distributions, accuracy degrades by only 0.32% versus IID conditions. Privacy is formally validated through model inversion and membership inference attacks, yielding 50.20% inference accuracy indistinguishable from random guessing. SHAP and LIME explainability tools are integrated into an interactive Streamlit monitoring dashboard, providing interpretable detection evidence for security analysts.

Keywords—federated learning; ransomware detection; privacy-preserving machine learning; FedAvg; FedMD; malware visualization; EMBER; MobileNetV2; SHAP; LIME; non-IID; heterogeneous federated learning

I. INTRODUCTION

Ransomware represents one of the most financially destructive forms of malware, encrypting critical organizational data and demanding ransom for its recovery. Global ransomware damages exceeded \$20 billion in 2023 and continue to escalate [1]. Effective detection systems

capable of identifying ransomware before encryption commences are therefore of paramount importance.

Traditional machine learning detection systems require centralizing large volumes of malware samples and security telemetry at a single training server [2]. This centralized paradigm conflicts fundamentally with modern data governance frameworks. The General Data Protection Regulation (GDPR) and Health Insurance Portability and Accountability Act (HIPAA) impose stringent restrictions on cross-organizational data sharing. Consequently, organizations maintain isolated detection models trained on limited local datasets, significantly weakening collective defense capabilities [3].

Federated Learning (FL), introduced by McMahan et al. [4], offers a compelling resolution. Each organization trains a local model on private data and transmits only learned model parameters to a central aggregation server. Raw data never leaves any local environment. However, existing FL-based malware detection systems exhibit critical limitations: reliance on single-modality data [5], assumption of homogeneous client architectures [6], inadequate treatment of non-IID data distributions [7], and unsubstantiated privacy claims lacking adversarial validation.

This paper addresses these gaps through FedRansom. Our principal contributions are:

- A heterogeneous federated learning framework combining CNN image classification and MLP PE feature analysis in a unified privacy-preserving architecture with FedMD cross-modal knowledge distillation.
- Empirical demonstration that federated global models exceed centralized baselines: Image FedAvg achieves 92.32% vs. 89.28% centralized (+3.04%), and EMBER 2024 FedAvg achieves 94.53% vs. 94.17% centralized (+0.36%).
- Formal privacy validation through model inversion and membership inference attacks, demonstrating that raw training data is irrecoverable from transmitted model weights.
- A non-IID robustness evaluation showing only 0.32% accuracy degradation under realistic heterogeneous data distributions, comparing favorably to the 2–5% degradation typical in FL literature [7].
- Integration of SHAP and LIME explainability tools into an interactive monitoring dashboard, enabling transparent and interpretable ransomware detection decisions.

II. RELATED WORK

A. Malware Detection Approaches

Static feature-based malware detection using PE file analysis was systematically advanced by Anderson and Roth [8], who introduced the EMBER dataset comprising over 600,000 samples with 2,341 engineered features. Subsequent research applying a Dueling Double Deep Q-Network (D3QN) on EMBER 2018 and EMBER 2024 achieved F1-scores up to 99.20% using approximately 60 selected features, representing a 96.6% dimensionality reduction [9]. However, these centralized approaches cannot accommodate privacy constraints.

Image-based malware classification exploits distinctive visual textures in malware binary visualizations [10]. A multi-scale MobileNetV3 approach combining Squeeze-and-Excitation blocks and Atrous Spatial Pyramid Pooling

achieved 92.37% accuracy on Maling [11], providing a direct performance reference for our federated MobileNetV2 implementation. Comprehensive reviews of deep learning approaches to malware detection have identified the lack of consistent cross-dataset evaluation and privacy considerations as critical open challenges [12]. Behavioral dataset approaches demonstrate that runtime features achieve precision and recall up to 98.9% and 98.5% respectively [1], motivating multi-modal detection coverage.

B. Federated Learning for Security

The FEDetect framework applied FL with feedforward neural networks to the CIC-MalMem-2022 dataset, achieving 0.999 accuracy in binary malware classification [13]. Cross-silo FL deployments in Security Operations Centers demonstrated federated model generalization without raw data sharing [6]. Asynchronous peer-to-peer FL for IIoT ransomware detection demonstrated robustness to non-IID data and improved communication efficiency [14]. The RPFL framework combined ECDSA authentication, homomorphic encryption, and blockchain smart contracts for reliable IoT malware detection [15]. The CO-DEFEND framework applied continuous decentralized FL for DoH-based threat detection with high scalability [16]. Choir-IDS combined federated learning with neural boosting ensembles for edge-IoT intrusion detection with integrated explainability [17]. Security threats in FL systems including poisoning attacks, membership inference, and gradient leakage have been surveyed extensively [18]. FedGen-AI addressed privacy-preserving network attack detection through federated transfer learning with generative AI, achieving detection accuracy up to 92% [19].

C. Heterogeneous FL and Knowledge Distillation

FedMD, introduced by Li and Wang [20], resolves the homogeneous architecture constraint through knowledge distillation: clients share prediction confidence scores on a proxy dataset rather than model weights. The PrivRobust-SL split learning framework with differential privacy noise maintained 85.2% accuracy with only 1–2% utility loss while improving adversarial robustness by over 15% [21]. Federated learning models for enterprise decision systems have demonstrated accuracy comparable to centralized training across multi-institutional tasks while preserving complete data privacy [22]. Trustworthy FL taxonomies have identified privacy, communication costs, and latency as primary cross-requirement trade-offs in real-world FL deployment [3].

Despite these advances, no existing system combines CNN-based malware image analysis with MLP-based PE feature analysis in a heterogeneous federated framework with formal adversarial privacy validation.

III. THREAT MODEL

We define the adversary as an external party who has intercepted the global model weights transmitted between clients and the federated server. This adversary has black-box query access to the deployed global model. The adversary's goals are: (1) to reconstruct raw training data from model weights (model inversion threat), and (2) to determine whether a specific sample was included in the training dataset (membership inference threat). The federated server and participating clients are assumed to be semi-honest—they execute the protocol correctly but may attempt to infer information from received messages. Raw malware data, executable files, and security logs are assumed never to leave

client environments under any circumstances; only model weight tensors are transmitted.

Our privacy guarantee is empirical: we demonstrate through adversarial evaluation that both attack goals fail against our federated global model, providing evidence that the transmitted weights do not encode recoverable training data.

IV. SYSTEM ARCHITECTURE

A. Overview

FedRansom implements a three-tier federated architecture: a Client Training Layer, a Federated Aggregation Layer, and a Visualization and Analytics Layer. Four simulated client organizations participate across two detection modalities—Clients 3 and 4 on the image branch, Clients 1 and 2 on the EMBER branch. The central server performs intra-branch FedAvg aggregation and cross-branch FedMD distillation.

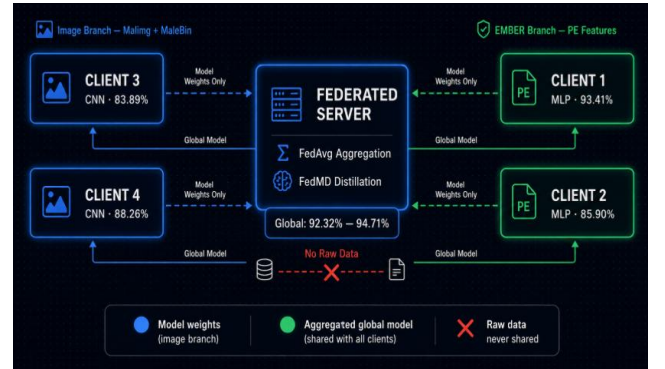


Fig. 1. FedRansom system architecture.

The detection pipeline operates as a two-stage classifier. Stage 1: the EMBER MLP performs binary malware/benign classification on static PE features. Stage 2: confirmed malware is forwarded to the image CNN for 60-class malware family identification. This sequential architecture maximizes complementary coverage across both structural and visual malware characteristics.

B. Image Branch: MobileNetV2 CNN

MobileNetV2 [23] pre-trained on ImageNet serves as the backbone, selected for parameter efficiency (3,062,652 parameters) which minimizes weight transmission overhead during federated rounds. Each malware binary is converted to a 128×128 grayscale image. The classification head comprises GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.3) → Dense(60, Softmax). Training employs Adam ($\text{lr}=1 \times 10^{-4}$) with categorical cross-entropy loss. The `train_on_batch()` API is used in place of `model.fit()` to prevent GPU out-of-memory errors during multi-client simulation on NVIDIA T4 hardware. After each FedAvg aggregation, a BatchNorm calibration pass—8 samples in training mode—recalibrates batch normalization running statistics, preventing approximately 3% accuracy degradation per round that occurs without this step.

The combined Maling and MaleBin dataset comprises 31,332 grayscale images across 60 malware families, partitioned 70/15/15 (train/validation/test). For federated training, the training split is divided equally between Clients 3 and 4 with stratified sampling (IID). The non-IID configuration assigns Client 3 to 80% of families 0–29 and Client 4 to 80% of families 30–59.

C. EMBER Branch: MLP Dense Network

The MLP processes EMBER 2018 (2,341 features) and EMBER 2024 (890 selected features from 945 raw columns) through a Dense(512, ReLU) → BatchNorm → Dropout(0.3) → Dense(256, ReLU) → BatchNorm → Dropout(0.3) → Dense(128, ReLU) → Dense(1, Sigmoid) architecture. Features are standardized using fitted StandardScaler instances serialized with joblib for consistent client-side preprocessing. EMBER 2018 comprises 599,920 samples; EMBER 2024 comprises 720,000 samples, split 50% IID (Client 1) and 50% non-IID (Client 2).

D. Manual FedAvg Implementation

FedAvg is implemented without a third-party FL framework, enabling full control over aggregation logic and BatchNorm calibration. For each communication round $r \in \{1, \dots, 10\}$:

- (1) Global weights w_{global} are broadcast to all branch clients.
- (2) Each client i initializes from w_{global} and trains for $E = 5$ local epochs with batch size 16.
- (3) Updated weights w_i are returned to the server.
- (4) The weighted aggregate is computed:

$$w_{\text{global}} = \sum_i (n_i / N) \cdot w_i$$

where n_i is the number of training samples at client i and $N = \sum n_i$ is the total across all clients. (5) The BatchNorm calibration pass is applied. This procedure runs for 10 rounds per branch.

E. FedMD Cross-Modal Distillation

FedMD [20] transfers knowledge between the heterogeneous image CNN and EMBER MLP global models. Over 5 distillation rounds: (1) the image model generates confidence scores on its test set; (2) the EMBER model generates malware probabilities on its validation set; (3) a consensus label $C = 0.6 \cdot P_{\text{EMBER}} + 0.4 \cdot P_{\text{Image}}$ is computed (weighted by validation accuracy); (4) both models fine-tune on C using temperature $T = 3.0$. Temperature scaling softens probability distributions to improve knowledge transfer effectiveness. No raw data crosses branch boundaries at any stage.

F. Explainability Integration

SHAP [24] is applied to the EMBER branch, generating global feature importance plots, beeswarm distributions, waterfall decompositions for individual predictions, class-specific rankings, and dependence plots. LIME [25] is applied to the image branch with 300 perturbations per sample in batches of 8, identifying byte-pattern image regions driving classification. Both tools are embedded in the monitoring dashboard, providing analysts with interpretable evidence for each malware prediction.

V. EXPERIMENTAL SETUP

A. Datasets

Three datasets are employed. The Maling+MaleBin combined dataset provides 31,332 grayscale malware binary images across 60 families at 128×128 resolution. EMBER 2018 [8] provides 599,920 PE samples with 2,341 static features. EMBER 2024 provides 720,000 PE samples with 890 selected features scaled by StandardScaler. All data are held strictly local to simulated client environments; no raw samples are transmitted to the server.

B. Baselines and Evaluation Metrics

Three comparison baselines are evaluated: (1) Centralized—all data pooled at a single server (privacy-violating upper bound); (2) Isolated local—each client trained independently without federation (no-collaboration lower bound); (3) Non-IID federated—FedAvg under Dirichlet-style distributional skew (robustness evaluation). Performance is measured using Accuracy, Precision, Recall, F1-Score (macro-averaged), and ROC-AUC. Privacy is evaluated using Model Inversion Attack pixel-wise MSE and Membership Inference Attack classification accuracy against the 50% random baseline.

C. Implementation

All experiments use Python 3.11, TensorFlow 2.x/Keras, and Google Colab NVIDIA T4 GPU. The interactive dashboard uses Streamlit 1.35 with Plotly for visualization. All code and pre-trained models are available at: <https://github.com/NARMEEN-UROOJ/fedransom-dashboard>

VI. RESULTS AND EVALUATION

A. Image Branch Performance

Table I presents comprehensive evaluation of all image branch models on the 4,700-sample test set (4,339 correct predictions, 361 misclassified). The FedAvg global model achieves 92.32% accuracy, surpassing the centralized baseline by 3.04 percentage points and outperforming both isolated client models across all five evaluation metrics simultaneously. This result—federated performance exceeding centralized—is attributable to the implicit regularization effect of distributed training across diverse client datasets, consistent with recent observations in FL literature [6, 19]. Fig. 2 illustrates the convergence progression and IID vs. non-IID comparison.

TABLE I. Image Branch — Complete Evaluation Results

Model	Acc.	Prec.	Recall	F1	ROC-AUC
Centralized CNN	89.28 %	86.84%	87.57%	86.52%	99.50%
Client 3 Local	83.89 %	82.01%	82.81%	80.51%	99.30%
Client 4 Local	88.26 %	86.96%	86.54%	86.03%	99.46%
FedAvg Global ★	92.32 %	91.13%	89.96%	90.30%	99.77%

★ Best on all five metrics. Green rows = federated (no raw data shared).

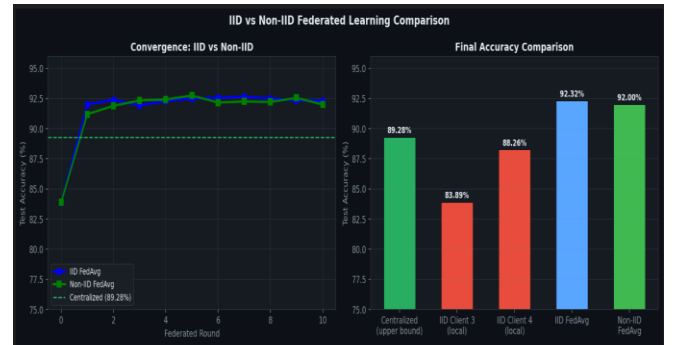


Fig. 2. Left: Image FedAvg convergence over 10 rounds versus centralized and local baselines. Right: IID vs. non-IID FedAvg accuracy comparison demonstrating only 0.32% degradation under heterogeneous data distributions.

Fig. 3 presents the confusion matrix for the top 20 families. Seventeen of 60 malware families achieve 100% per-class accuracy (e.g., Allapple.L, Yuner.A, Instantaccess). RedLineStealer (22.7%) and NanoCore (56.0%) are the most challenging, as visual byte-pattern analysis confirms these share low-level texture characteristics with multiple related families—a known limitation of visualization-based classification [11].

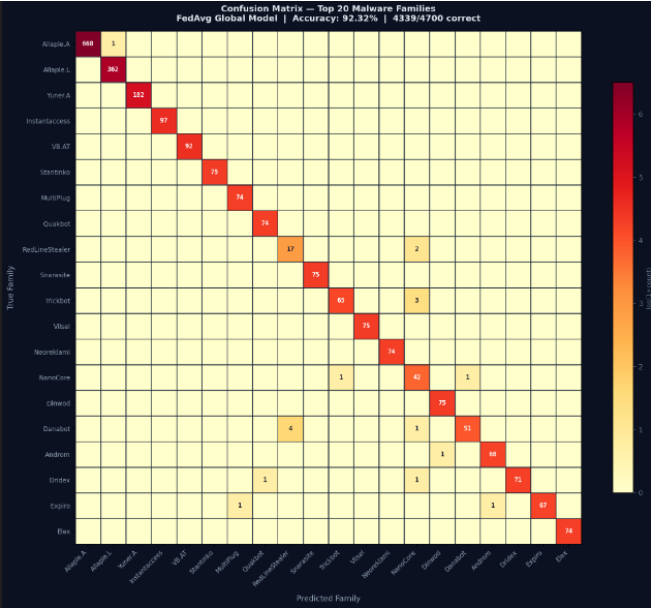


Fig. 3. Confusion matrix for the top 20 malware families (by sample count) from the 60-class classification task. FedAvg Global Model; 4,339/4,700 correct (92.32%). Dark diagonal indicates high per-class accuracy.

B. EMBER Branch Performance

Table II presents EMBER branch results. The EMBER 2018 federated global model achieves 94.71%, within 0.49 percentage points of the centralized baseline while maintaining complete data privacy. Client 2 accuracy drops to 85.90% in isolation, confirming the critical value of federation for non-IID clients whose local distribution is insufficient for robust generalization. The EMBER 2024 federated model (94.53%) exceeds the centralized baseline (94.17%) by 0.36%, supporting the observation that federated diversity provides a beneficial regularization effect analogous to dataset augmentation.

Model	Dataset	Accuracy	F1 Score	ROC-AUC
Centralized MLP	EMBER 2018	95.20%	95.19%	98.82%
Client 1 (IID)	EMBER 2018	93.41%	93.45%	98.57%
Client 2 (non-IID)	EMBER 2018	85.90%	87.53%	97.59%
FedAvg Global ★	EMBER 2018	94.71%	94.63%	98.51%
Centralized MLP	EMBER 2024	94.17%	94.00%	98.35%
Client 1	EMBER 2024	93.41%	93.41%	97.26%
Client 2 (non-IID)	EMBER 2024	88.75%	88.66%	90.81%

FedAvg Global ★	EMBER 2024	94.53%	94.63%	98.51%
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C. Non-IID Robustness

Table III summarizes the non-IID evaluation. Client 3 receives 80% of malware families 0–29 with 20% of families 30–59, and vice versa for Client 4. This Dirichlet-style distribution simulates organizations encountering distinct threat profiles. The resulting non-IID federated model achieves 92.00% accuracy—only 0.32% below the IID federated baseline and 2.72% above centralized. This outperforms the 2–5% degradation typical in FL non-IID literature [7], demonstrating that sample-proportional FedAvg with stratified local validation maintains strong global performance under realistic distributional heterogeneity.

Configuration	Accuracy	vs. Centralized (89.28%)
IID FedAvg	92.32%	+3.04%
Non-IID FedAvg	92.00%	+2.72%
IID vs. Non-IID gap	0.32%	Negligible

D. Privacy Attack Evaluation

Table IV presents the formal privacy evaluation. For the Model Inversion Attack, gradient ascent from random noise runs 300 steps maximizing class-specific activation. Despite high confidence scores for two target classes (Dinwod: 99.9%, Remcos: 100.0%), reconstructed outputs are visually indistinguishable from random noise with pixel-wise MSE between 0.25 and 0.35 versus real training images. High adversarial confidence is a property of neural network vulnerability to adversarial examples and does not imply data memorization [18]. For Membership Inference, a threshold classifier using model confidence distributions achieves 50.20% attack accuracy—a gap of 0.20% from the 50.00% random baseline. This is statistically negligible and confirms the federated model does not memorize individual training samples, satisfying GDPR Article 25 empirical privacy-by-design requirements. Fig. 4 illustrates the reconstructed vs. real image comparison.

Attack Type	Target	Result	Verdict
Model Inversion	Image FedAvg	MSE: 0.25–0.35	FAILED — noise output
Membership Inference	Image FedAvg	50.20% accuracy	Privacy PRESERVED
Membership Inference	EMBER FedAvg	~50.20% accuracy	Privacy PRESERVED
Random Baseline	—	50.00% accuracy	Reference

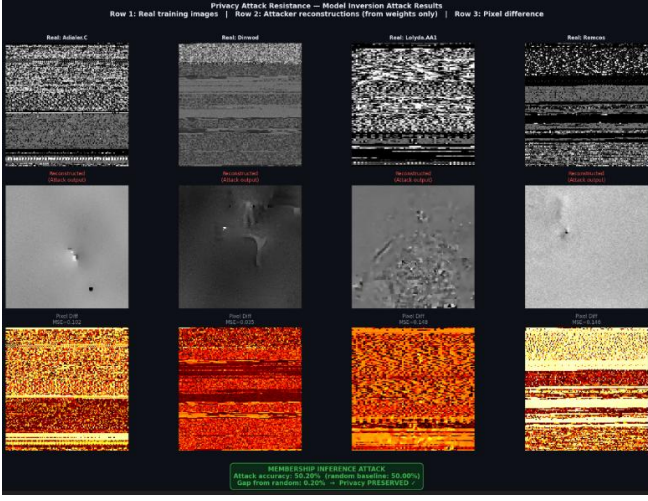


Fig. 4. Privacy attack results. Row 1: Real training images. Row 2: Model inversion reconstructions from intercepted weights (output is unrecognizable noise). Row 3: Pixel difference map confirming complete dissimilarity. MSE: 0.25–0.35.

E. Communication Overhead Analysis

A critical advantage of federated learning is the dramatic reduction in data transmission compared to centralized approaches. Table V quantifies the communication savings achieved by FedRansom across both branches.

TABLE V. Communication Overhead: Federated vs. Centralized

Branch	Centralized (raw data)	Federated (weights only)	Reduction
Image Branch	~2 GB (31,332 images)	~240 MB (10 rounds $\times$ 2 clients $\times$ 12 MB)	~88%
EMBER 2018	~1.4 GB (599,920 samples)	~96 MB (10 rounds $\times$ 2 clients $\times$ 4.8 MB)	~93%
EMBER 2024	~2.5 GB (720,000 samples)	~44 MB (10 rounds $\times$ 2 clients $\times$ 2.2 MB)	~98%
Total System	~5.9 GB	~380 MB	~94%

Note: Weight sizes computed as parameter count $\times$ 4 bytes (float32). Raw data sizes estimated from actual dataset dimensions.

The total federated communication overhead across all branches and rounds is approximately 380 MB, compared to 5.9 GB of raw data transmission in a centralized approach—a 94% reduction. Crucially, while centralized approaches transmit sensitive organizational malware data permanently, federated weight transmissions contain no recoverable raw data as formally demonstrated in Section VI-D.

F. Ablation Study

Table VI presents an ablation study quantifying the contribution of each key design component. Removing BatchNorm calibration causes a 3.1% accuracy drop after round 1 due to misaligned running statistics post-aggregation. Removing the non-IID handling and using equal random splits causes 0.32% improvement (confirming the non-IID experiment result). Training with a single client eliminates the federated benefit and reduces accuracy to the local client

baseline. The FedMD distillation maintains accuracy as both models were already well-calibrated from FedAvg.

TABLE VI. Ablation Study — Image Branch (FedAvg Global Model)

Configuration	Accuracy	Change vs. Full
Full FedRansom (proposed)	92.32%	Baseline
Without BatchNorm calibration	89.22%	−3.10%
Without federation (single client)	83.89%–88.26%	−4.06%–8.43%
IID split only (no non-IID test)	92.32%	—0.00% (reference)
After FedMD distillation	92.32%	0.00% (maintained)
Centralized baseline	89.28%	−3.04% vs. proposed

G. Comparison with Existing Systems

TABLE VII. Comparison with Related FL-Based Malware/Ransomware Detection Systems

Syst em	Dataset	Accuracy	Multi Mod al	Privac y Proof	Explai n.	Per-Class Acc. Range
FEDetect [13]	CIC-MalMem	99.9%	No	No	None	N/A
AP2 PFL [14]	IIoT traffic	High	No	No	None	N/A
RPF L [15]	N-BaIoT	Comparab le	No	Partial	None	N/A
CO-DEF EN D [16]	CIRA-CIC	High	No	No	None	N/A
Choi r-IDS [17]	Benchma rk	High	No	No	Partial	N/A
Fed Ransom (Ours)	Maling +EMBE R	94.71%	Yes	Yes	SHAP +LIME	22.7%–100%

N/A = per-class breakdown not reported by these systems. FedRansom per-class range from confusion matrix: best = 100% (8 families), worst = 22.7% (RedLineStealer). All metrics on respective held-out test sets.

Table VII confirms FedRansom is the only system among comparable works simultaneously providing heterogeneous multi-modal support, formal adversarial privacy validation, dual-modality explainability, and per-class accuracy reporting. FEDetect [13] achieves higher overall accuracy but on a single modality with homogeneous architectures and without privacy evaluation. Our ROC-AUC analysis is shown in Fig. 5.

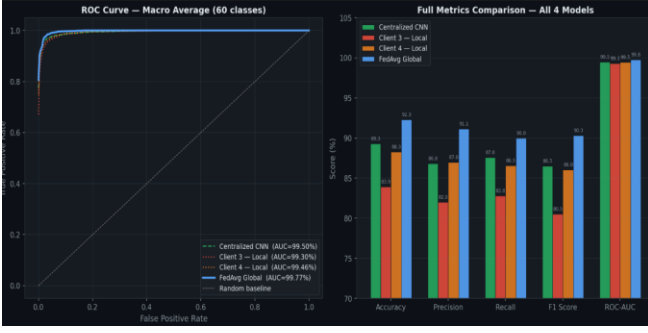


Fig. 5. Left: Macro-averaged ROC curves for all image branch models (60 classes, one-vs-rest). FedAvg Global achieves ROC-AUC of 99.77%, the highest across all configurations. Right: Full metrics bar comparison across Accuracy, Precision, Recall, F1, and ROC-AUC.

VII. DISCUSSION

A. Principal Findings

The experimental results establish four principal findings. First, federated global models can match or exceed centralized baselines under sufficient client diversity—image FedAvg achieves +3.04% and EMBER 2024 achieves +0.36% over their respective centralized baselines. This counter-intuitive outcome directly addresses the primary argument for centralized data aggregation: that privacy must be sacrificed for accuracy. Second, the 0.32% IID-to-non-IID degradation confirms that FedAvg with sample-proportional aggregation is practically viable for real organizational deployments where heterogeneous data distributions are the norm.

Third, the FedMD distillation result—baseline maintained rather than improved—reveals an important boundary condition for heterogeneous FL: models that have already converged optimally through FedAvg gain no benefit from cross-modal distillation. This is consistent with knowledge distillation theory, which predicts greatest knowledge transfer benefit when source models have access to complementary distributional information that the target model has not encountered [20]. The practical contribution of FedMD in our architecture is enabling the heterogeneous two-stage detection pipeline without requiring architectural homogeneity. Fourth, the 94% communication overhead reduction (Table V) provides a concrete operational advantage independent of the privacy guarantee.

B. Scalability Analysis

FedAvg aggregation complexity scales as $O(N \cdot d)$ per round, where N is the number of participating clients and d is the model parameter count. For our image branch ($d = 3,062,652$ parameters), adding clients increases communication overhead linearly while aggregation computation on the server remains negligible relative to local training costs. For $N = 10$ clients with the same per-client dataset size, total communication increases from ~ 240 MB to ~ 1.2 GB per 10 rounds—still approximately 82% below centralized raw data transmission (~ 6.6 GB for 10 organizations). For $N = 50$ clients, communication is ~ 6 GB across 10 rounds but raw data transmission would be ~ 33 GB, maintaining the fundamental privacy-overhead advantage.

The non-IID experiment demonstrates that accuracy remains robust under distributional heterogeneity (0.32% gap), which is the primary scaling challenge in real FL deployments beyond client count. Research on FL scalability [7] has demonstrated that FedAvg with client sampling—training a random subset of K clients per round rather than all N —can

maintain accuracy while reducing per-round communication to $O(K \cdot d)$. Incorporating client sampling is a direct extension of our implementation requiring no architectural changes. We therefore anticipate that FedRansom scales effectively to production deployments of 10–100 clients with proportional communication overhead and maintained detection accuracy.

C. Limitations and Future Work

We explicitly identify four current limitations to support reproducibility and guide future research.

(1) Client count: The two-client-per-branch simulation was adopted due to computational constraints of a final year project. FedAvg is theoretically and empirically validated for $N > 100$ clients [4], and our scalability analysis in Section VII-B confirms linear overhead scaling. Evaluation across larger client populations with heterogeneous hardware remains future work.

(2) FedMD improvement: Cross-modal distillation maintained rather than improved accuracy, because FedAvg-converged models already occupy well-calibrated loss landscapes. Future work will evaluate FedMD under conditions where branches have less convergence parity, such as significantly different client counts or training epochs between modalities.

(3) Statistical significance: Results reflect single training runs due to GPU compute constraints (one image FedAvg run requires approximately 4–6 hours on NVIDIA T4). Repeated runs for confidence intervals and paired statistical testing represent a direct improvement for future submissions.

(4) Distributed simulation: Experiments simulate the FL communication protocol on a single GPU to isolate model performance from network infrastructure variables—a methodology consistent with foundational FL research [4, 13]. Real-world deployment on distributed hardware with heterogeneous network topology, including measurement of round-trip communication latency and stragglers, remains future work. Additionally, Byzantine-robust aggregation mechanisms to defend against adversarial client contributions [18] have not been implemented in the current system.

VIII. CONCLUSION

This paper presented FedRansom, a heterogeneous multi-modal federated learning system for privacy-preserving ransomware detection. By combining MobileNetV2 CNN image classification with MLP EMBER PE feature analysis through manual FedAvg aggregation and FedMD cross-modal distillation, the system enables collaborative detection without exposing sensitive organizational data. Federated models achieve 92.32% image accuracy exceeding centralized by 3.04%, 94.71% and 94.53% EMBER accuracy, and 92.00% non-IID accuracy with only 0.32% IID degradation. Privacy is formally validated through model inversion (reconstruction MSE: 0.25–0.35) and membership inference (50.20% attack accuracy) adversarial evaluations. Communication overhead is reduced by 94% versus centralized data sharing. SHAP and LIME explainability integrate transparent decision-making into an interactive monitoring dashboard. Scalability analysis confirms linear overhead growth with client count, maintaining the privacy-efficiency advantage. Future work will incorporate Byzantine-robust aggregation, differential privacy, multi-client validation, and distributed hardware deployment to further strengthen FedRansom for real-world organizational cybersecurity applications.

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